AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

AQA A-level Chemistry · Revision questions · Sheet 9 of 9

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- K_c: divide each amount by the volume in dm^3 before substituting (unless the volumes cancel)
- Only temperature changes the value of K_p or K_c; pressure, concentration and catalysts do not
- Exothermic forward reaction: raising temperature lowers K; endothermic: raising temperature increases K
- Partial pressure = mole fraction × total pressure
- K_p uses partial pressures in round brackets, e.g. $K_p = p(C)^2 \div p(A)\ p(B)^3$; never square brackets

Section A: Short-answer questions

01 The forward reaction of an equilibrium is exothermic. State the effect of increasing the temperature on the value of K_p.

[1 mark]

02 State why a dynamic equilibrium can only be established in a closed system.

[1 mark]

03 Which statement is **not** always correct for a reaction at equilibrium? (reactants ⇌ products)

[1 mark]

A The concentrations of the reactants and products are equal.
B The equilibrium can be achieved starting from the reactants.
C The equilibrium can be achieved starting from the products.
D The rate of the forward reaction is equal to the rate of the reverse reaction.

Section B: Longer questions

04 1 mol of a diester with molecular formula $C_7H_{12}O_4$ is added to 1 mol of water in the presence of a small amount of catalyst. The mixture is left to reach equilibrium at a constant temperature.

$C_7H_{12}O_4(l) + 2H_2O(l) \rightleftharpoons 2CH_3COOH(l) + HO(CH_2)_3OH(l)$

04.1 At equilibrium, *x* mol of ethanoic acid are present in the mixture. Complete the table by deducing the amounts, in terms of *x*, of the diester, water and diol present in the equilibrium mixture.

Amount / mol	Diester	Water	Acid	Diol
At the start	1	1	0	0
At equilibrium			*x*	

[3 marks]

04.2 A new equilibrium mixture is prepared at a different temperature. The table shows the amounts in this new equilibrium mixture.

Amount / mol	Diester	Water	Acid	Diol
At equilibrium	0.971	To be calculated	0.452	0.273

The value of the equilibrium constant, K_c, is 0.161 at this temperature. Calculate the amount of water, in mol, in this new equilibrium mixture. Show your working.

[3 marks]

05 This question is about the equilibrium

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

05.1 State and explain the effect, if any, of a decrease in overall pressure on the equilibrium yield of SO_3.

[3 marks]

05.2 A 0.460 mol sample of SO_2 is mixed with a 0.250 mol sample of O_2 in a sealed container at a constant temperature. When equilibrium is reached at a pressure of 215 kPa, the mixture contains 0.180 mol of SO_3. Calculate the partial pressure, in kPa, of SO_2 in this equilibrium mixture.

[4 marks]

05.3 A different mixture reaches equilibrium at a different temperature. The table shows the partial pressures of the gases at equilibrium.

Gas	Partial pressure / kPa
SO_2	1.67×10^2
O_2	1.02×10^2
SO_3	1.85×10^2

Give an expression for K_p for this reaction. Calculate the value of K_p and give its units.

[3 marks]

05.4 State the effect on the value of K_p if the pressure of this equilibrium mixture is increased at a constant temperature.

[1 mark]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	K_p decreases	*Source: Practice question (written for this set)*	1
02	So that no reactants or products can escape	*Source: Practice question (written for this set)*	1
03	**A**	*Source: 7404/1 Nov 2020 Q19*	1
04.1	M1 diester = $1 - x/2$ M2 water = $1 - x$ M3 diol = $x/2$	*Source: 7405/2 Nov 2021 Q05.1, 05.3*	3
04.2	M1 $K_c = 0.452^2 \times 0.273 \div (0.971 \times (\text{amount } H_2O)^2)$ M2 $(\text{amount } H_2O)^2 = 0.452^2 \times 0.273 \div (0.161 \times 0.971)$ (= 0.357) M3 amount $H_2O = \sqrt{0.357} = 0.597$ mol	Volume cancels	3
05.1	M1 decreases yield M2 more moles / molecules on the left-hand side M3 so equilibrium shifts (to left) to oppose the decrease in pressure / to increase pressure	*Source: 7405/1 Nov 2021 Q05* Must refer to equilibrium shifting for full marks	3
05.2	M1 $n(SO_2) = 0.46 - 0.18 = 0.28$ mol M2 $n(O_2) = 0.25 - 0.09 = 0.16$ mol M3 total = 0.62 mol M4 $p(SO_2) = 0.28 \div 0.62 \times 215 = 97(.1)$ kPa		4
05.3	M1 $K_p = p(SO_3)^2 \div (p(SO_2)^2 \times p(O_2))$ M2 $= 1.2(0) \times 10^{-2}$ M3 kPa^{-1}	Penalise square brackets in M1	3
05.4	Stays the same	Original is tick-one-box	1